

TOURIST ATTRACTION PLANNER

Your preferences. Our algorithms. One optimized journey.

“An intelligent planner that selects and orders tourist attractions based on time, budget and personal preferences.”

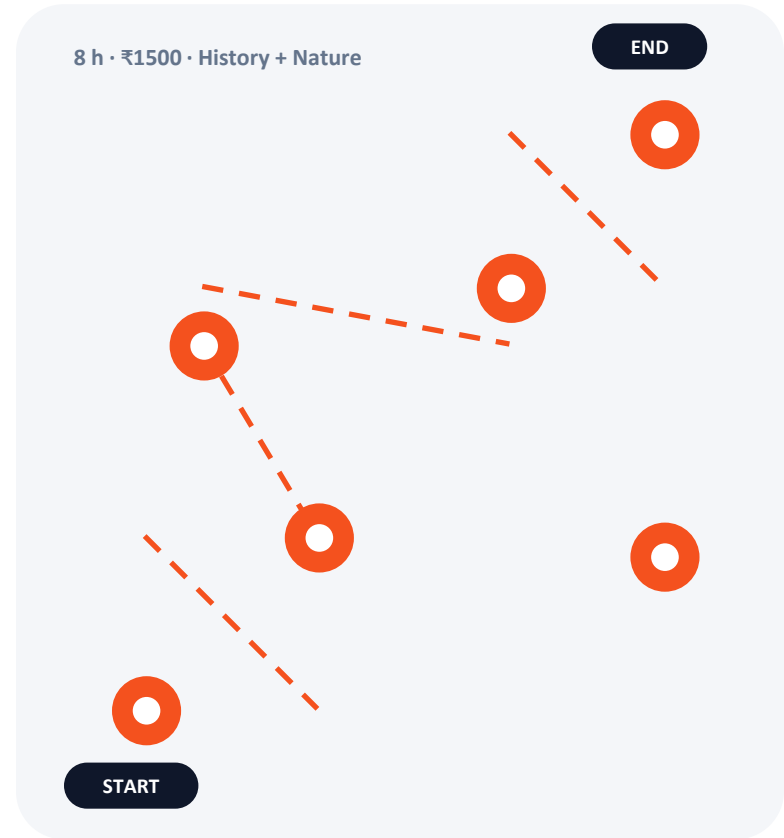
PROBLEM DOMAIN: Travel Planning & Optimization

TEAM MEMBERS : B.AMRUTHA VARSHINI

M.SWATHI

R.RAMYASRI

M.BHAVANA



The Challenge

Planning manually gets hard with limited time, limited budget, many attractions, different interests and travel time between places .



Time constraint



Budget constraint



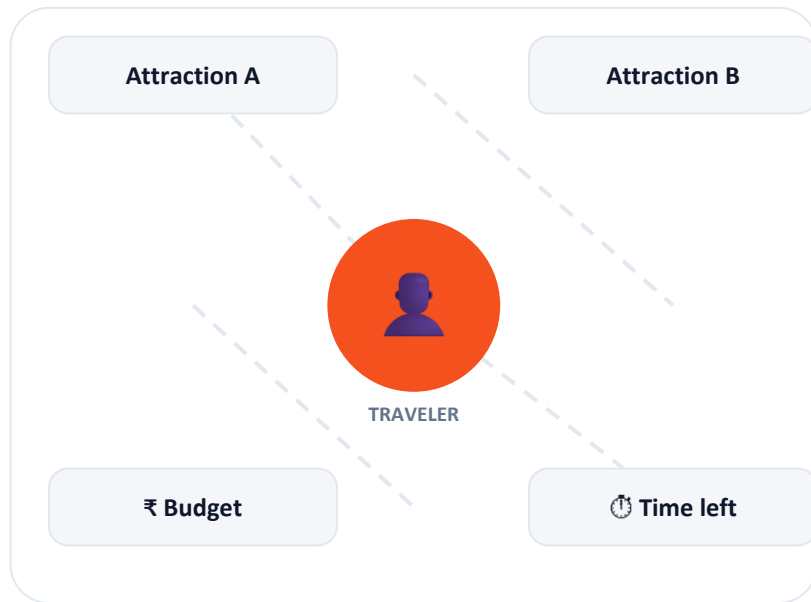
Travel distance



Personal preferences



Changing trip conditions



“How can we create the most valuable feasible trip within real-world constraints?”

Our Innovation

AI-based preference understanding + classical optimization = a personalized, explainable travel plan.



Personalized Scoring

Understands what the traveler values



Greedy Optimization

Quickly picks high-value feasible stops



Knapsack Optimization

Best combination under time & budget



Graph-Based Routing

Attractions and travel links as a graph



Dynamic Replanning

Adapts when time, budget or places change



History Explorer

Historical & cultural context for places

AI Preferences



Optimization



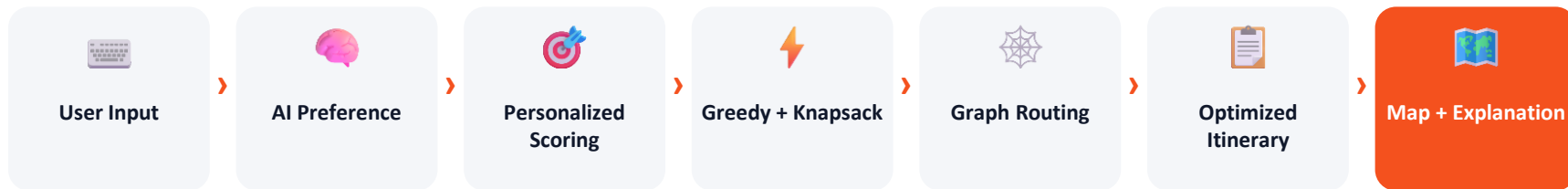
Route



Explainable Trip

AI understands the traveler. Algorithms optimize the trip. The system explains every decision.

How It Works



USER JOURNEY



INPUT

"8 hours + ₹1500 + History + Nature"



OUTPUT

Personalized attractions + optimized route + budget/time summary

Technology & Algorithms

1 · GREEDY



Picks the highest personalized value-per-resource attraction first.

WHY Fast; shows real-time decisions

$O(n \log n)$ after sorting



2 · 0/1 KNAPSACK



Picks the best combination within time and budget limits.

WHY Suited to constrained selection

$O(n \times T \times B)$ discretized DP



3 · GRAPH TRAVERSAL



Attractions are nodes; travel links are weighted edges.

WHY Explores connectivity, builds routes

BFS/DFS: $O(V + E)$

Python

Streamlit

Pandas

NetworkX

Folium / Leaflet

Plotly

Demo & Impact

Trip constraints

Time available (hours)

Budget (₹)

Starting location

Preferred travel style

Relaxed
Balanced
Packed

What do you enjoy?

Select all that apply

History
Nature
Culture
Food
Architecture
Shopping
Photography
Family

Describe your ideal trip

I have 8 hours and ₹1500. I love history, photography and nature.

▼ AI understood your preferences (local rule-based NLP)

History	High	Nature	High
Culture	Medium	Architecture	Medium
Photography	High	Budget focus	Medium
Time focus	Medium		

8h ₹1500 Balanced Detected "Photography" in text Time → 8h Budget → ₹1500

SAMPLE OUTPUT

Selected Attractions: 5
 Budget Used: ₹920 / ₹1500
 Time Used: 6h 40m / 8h
 Start → A → B → C → D → End

- ✓ Reduces manual planning
- ✓ Makes optimization understandable
- ✓ Adapts to changing constraints
- ✓ Personalized recommendations
- ✓ Explains why places are selected or skipped

MANUAL PLANNING

Multiple constraints + trial and error



TRIOPTIMA AI

Personalization + Optimization + Route + Explanation

Future Vision

PROTOTYPE TODAY

- ✓ Personalized scoring
- ✓ Greedy optimization
- ✓ Knapsack optimization
- ✓ Graph traversal
- ✓ Route visualization
- ✓ What-If simulation
- ✓ Dynamic replanning
- ✓ Historical information

NEXT

-  Real-time map & traffic
-  Weather-aware planning
-  Congestion-aware routes
-  Expanded history
-  Advanced AI preferences
-  Mobile-first deployment

SCALABILITY From a demo dataset to city-level and multi-city planning systems.

“IT turns travel planning into an explainable optimization problem.”

THANK YOU · Q&A